```
!nvidia-smi
```

```
Tue Feb 24 07:29:02 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.105.08             Driver Version: 580.105.08     CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA L4                      Off |   00000000:00:03.0 Off |                    0 |
| N/A   43C    P8             13W /   72W |       0MiB /  23034MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
```

```
import torch

torch.cuda.is_available()
```

```
True
```

Data Acquisition

```
%%bash

wget https://raw.githubusercontent.com/inllclj/Project-1274287N/main/defect_cast.zip
wget https://raw.githubusercontent.com/inllclj/Project-1274287N/main/ok_cast.zip


```

```
!ls -al
```

```
total 164260
drwxr-xr-x 6 root root      4096 Feb 24 07:30 .
drwxr-xr-x 1 root root      4096 Feb 24 07:23 ..
-rw-r--r-- 1 root root 136032048 Feb 23 17:56 best_casting_model.h5
drwxr-xr-x 5 root root      4096 Feb 23 08:07 data
drwxr-xr-x 4 root root      4096 Jan 28 05:23 dataset
-rw-r--r-- 1 root root  20078407 Feb 24 07:29 defect_cast.zip
drwxr-xr-x 2 root root      4096 Jan 27 15:22 .ipynb_checkpoints
```

```
-rw-r--r-- 1 root root  12065685 Feb 24 07:30 ok_cast.zip
drwxr-xr-x 3 root root      4096 Feb 23 16:32 SimpleCNN
```

Unzip images

```
%%bash
unzip -o ok_cast.zip -d data
unzip -o defect_cast.zip -d data
```

Data Understanding & Exploration (EDA)

- Inspect dataset structure and class labels (defective vs non-defective).
- Count images per class to identify class imbalance.
- Visually inspect sample images to understand:

- Defect appearance

- Lighting conditions

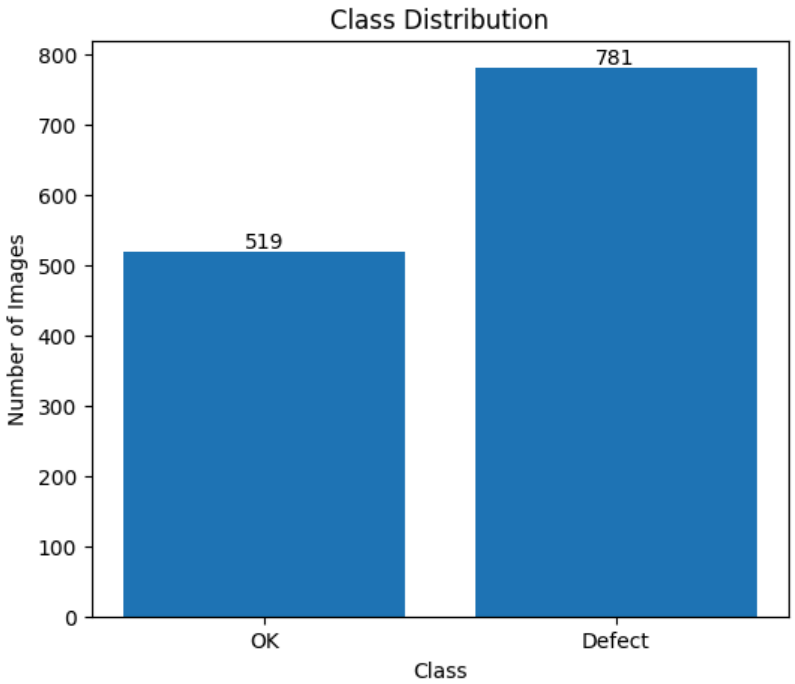
- Background consistency

- Image resolution and color format

```
# Class dstribution
import os
import matplotlib.pyplot as plt

# Paths
ok_dir = "data/ok_cast"
defect_dir = "data/defect_cast"

# Count images
ok_images = [f for f in os.listdir(ok_dir) if f.lower().endswith('.jpeg')]
defect_images = [f for f in os.listdir(defect_dir) if f.lower().endswith('.jpeg')]

ok_count = len(ok_images)
defect_count = len(defect_images)

print("Class Distribution:")
print("-------------------")
print(f"OK Images:  {ok_count}")
print(f"Defect Images: {defect_count}")

classes = ['OK', 'Defect']
```

```
counts = [ok_count, defect_count]

plt.figure(figsize=(6,5))
plt.bar(classes, counts)
plt.title("Class Distribution")
plt.xlabel("Class")
plt.ylabel("Number of Images")

# Show values on bars
for i, v in enumerate(counts):
    plt.text(i, v + 5, str(v), ha='center')

plt.show()
```

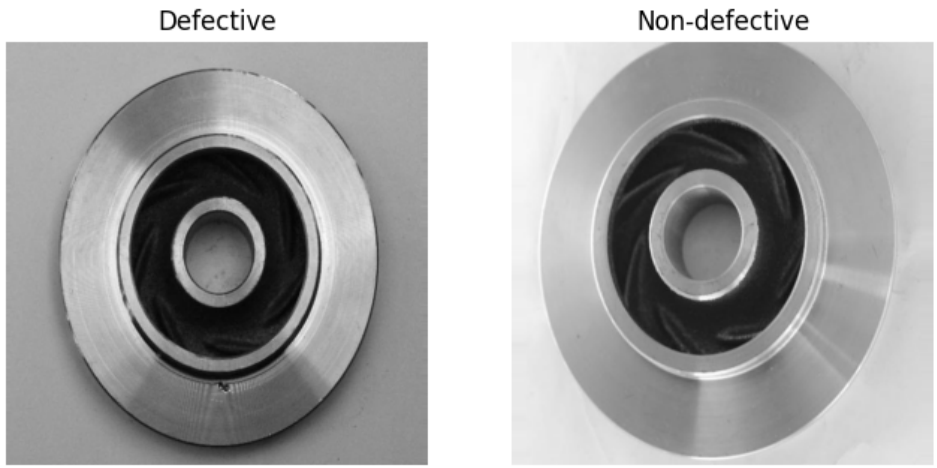
```
Class Distribution:
-------------------
OK Images:  519
Defect Images: 781
```




```
# Visual inspection
import os
import random
import matplotlib.pyplot as plt
from PIL import Image

```

```python
fig, axes = plt.subplots(1, 2, figsize=(8, 4))

# Defective image
random_defect_img = Image.open(os.path.join("data/defect_cast", random.choice(
    [f for f in os.listdir("data/defect_cast") if f.endswith('.jpeg')]
    )))
axes[0].imshow(random_defect_img)
axes[0].set_title("Defective")
print(f"Defect Images size: {random_defect_img.size}")
axes[0].axis("off")

# OK image
random_ok_img = Image.open(os.path.join("data/ok_cast", random.choice(
    [f for f in os.listdir("data/ok_cast") if f.endswith('.jpeg')]
    )))
axes[1].imshow(random_ok_img)
axes[1].set_title("Non-defective")
print(f"OK Images size:  {random_ok_img.size}")
axes[1].axis("off")

plt.show()

```

```
Defect Images size: (512, 512)
OK Images size:  (512, 512)
```




```python
# Data Cleaning, remove corrupted images
from PIL import Image
import os

def remove_corrupted_images(folder):
    for img_name in os.listdir(folder):
        img_path = os.path.join(folder, img_name)
```

```python

        # Skip non-image files
        if not img_name.lower().endswith('.jpeg'):
            continue

        try:
            with Image.open(img_path) as img:
                img.verify()  # Check if image is corrupted
        except Exception:
            print(f"Removing corrupted image: {img_name}")
            os.remove(img_path)

base_dir = "data/defect_cast"
remove_corrupted_images(base_dir)

base_dir = "data/ok_cast"
remove_corrupted_images(base_dir)


```

```python
import os
import shutil
from sklearn.model_selection import train_test_split

def split_data(src, train_dir, test_dir, split=0.8):
    """
    Clears target directories and splits images from src into train/test sets.
    """

    # 1. Clean up or create directories
    for directory in [train_dir, test_dir]:
        if os.path.exists(directory):
            print(f"Cleaning existing directory: {directory}")
            # Remove all files in the directory without deleting the folder itself
            shutil.rmtree(directory)

        os.makedirs(directory, exist_ok=True)

    # 2. Gather files
    # Lowercase check ensures we catch .JPEG, .jpg, and .jpeg
    all_images = [f for f in os.listdir(src)
                  if f.lower().endswith(('.jpeg', '.jpg'))]

    if not all_images:
        print(f"Warning: No images found in {src}")
        return

    # 3. Perform the split
    train_imgs, test_imgs = train_test_split(
```

```
        all_images,
        test_size=1 - split,
        random_state=42
    )

    # 4. Copy files to new locations
    print(f"Copying {len(train_imgs)} images to {train_dir}...")
    for img in train_imgs:
        shutil.copy(os.path.join(src, img), os.path.join(train_dir, img))

    print(f"Copying {len(test_imgs)} images to {test_dir}...")
    for img in test_imgs:
        shutil.copy(os.path.join(src, img), os.path.join(test_dir, img))

    print("Done!")
```

Preparing the data

Split the data into train and test by 80 to 20 proportion for ok images and def images

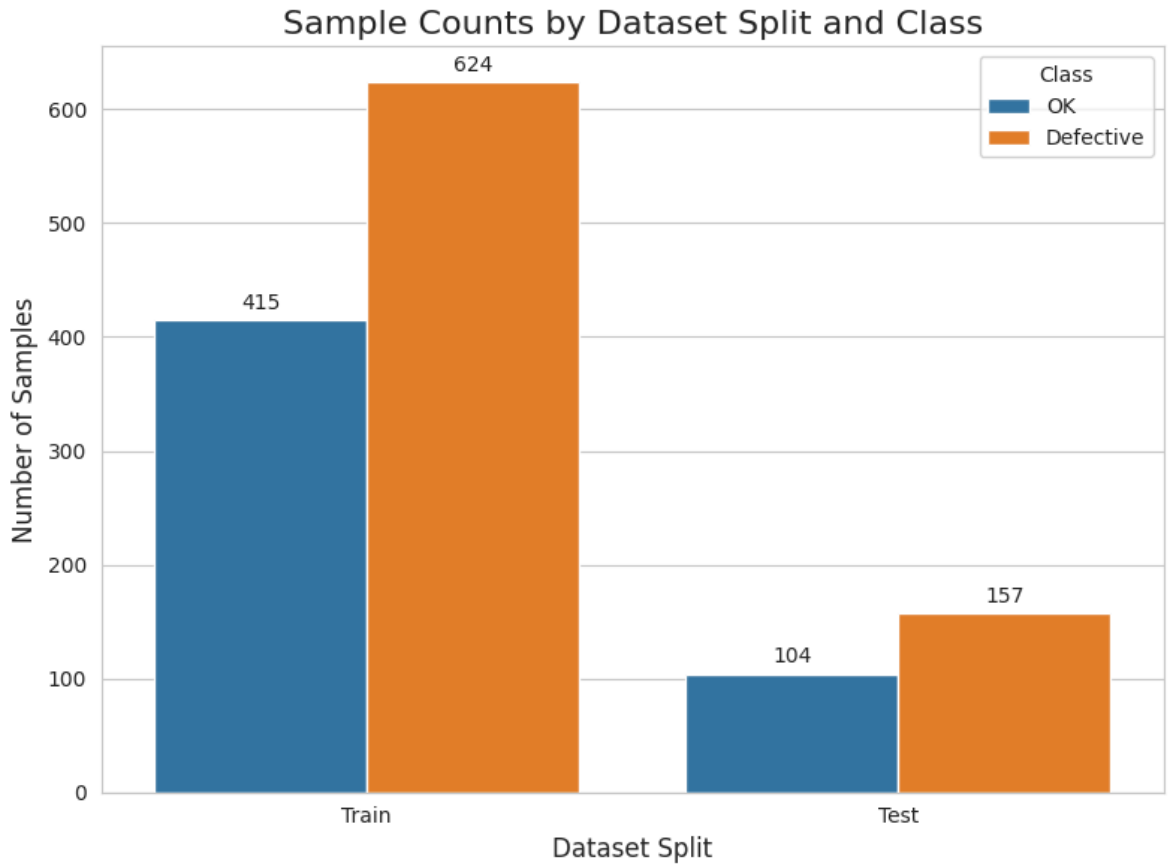
```
# OK images
split_data(
    src="data/ok_cast",
    train_dir="dataset/train/ok",
    test_dir="dataset/test/ok",
    split=0.8
)

# Defective images
split_data(
    src="data/defect_cast",
    train_dir="dataset/train/faulty",
    test_dir="dataset/test/faulty",
    split=0.8
)
```

```
Copying 415 images to dataset/train/ok...
Copying 104 images to dataset/test/ok...
Done!
Copying 624 images to dataset/train/faulty...
Copying 157 images to dataset/test/faulty...
Done!
```

```
import os
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

```

```
# ----------------------------
# Directory configuration
# ----------------------------
ROOT_DIR = "/content/dataset"
TRAIN_DIR = os.path.join(ROOT_DIR, "train")
TEST_DIR = os.path.join(ROOT_DIR, "test")

CLASS_NAMES = ["ok", "faulty"]

# ----------------------------
# Helper function
# ----------------------------
def count_images(base_dir, class_name):
    class_path = os.path.join(base_dir, class_name)
    if not os.path.exists(class_path):
        return 0
    return len([
        f for f in os.listdir(class_path)
        if f.lower().endswith(".jpeg")
    ])

# ----------------------------
# Count samples
# ----------------------------
train_ok = count_images(TRAIN_DIR, "ok")
train_defect = count_images(TRAIN_DIR, "faulty")
test_ok = count_images(TEST_DIR, "ok")
test_defect = count_images(TEST_DIR, "faulty")

# ----------------------------
# Create DataFrame
# ----------------------------
df = pd.DataFrame({
    "Split": ["Train", "Train", "Test", "Test"],
    "Class": ["OK", "Defective", "OK", "Defective"],
    "Count": [train_ok, train_defect, test_ok, test_defect]
})

# ----------------------------
# Visualization
# ----------------------------
sns.set_style("whitegrid")
plt.figure(figsize=(8, 6))

ax = sns.barplot(
    data=df,
    x="Split",
    y="Count",
    hue="Class"
)
```

```python

ax.set_title("Sample Counts by Dataset Split and Class", fontsize=16)
ax.set_xlabel("Dataset Split", fontsize=12)
ax.set_ylabel("Number of Samples", fontsize=12)

# Add value labels
for container in ax.containers:
    ax.bar_label(container, padding=3)

plt.tight_layout()
plt.show()

# ----------------------------
# Class ratio reporting
# ----------------------------
if train_defect > 0:
    print(f"Train ratio (OK / Defective): {train_ok / train_defect:.2f}")
else:
    print("Train ratio: Defective class missing")

if test_defect > 0:
    print(f"Test ratio (OK / Defective): {test_ok / test_defect:.2f}")
else:
    print("Test ratio: Defective class missing")
```




```
Train ratio (OK / Defective): 0.67
Test ratio (OK / Defective): 0.66
```

The code counts the number of images in each class for training and testing datasets, stores the results in a DataFrame, visualizes the class distribution using a bar chart, and calculates the ratio between OK and defective samples to assess class balance.

This code helps to:

-Detect class imbalance

-Verify dataset split correctness

-Visualize distribution

-Avoid biased training

```
!pip install mlflow
```

```python
import torch
import random
from glob import glob
from os.path import dirname, basename
from torch.utils.data import Dataset, DataLoader
from torchvision import transforms
from torchvision.io import read_image, ImageReadMode

device = "cuda" if torch.cuda.is_available() else "cpu"

class CastingDataset(Dataset):
    def __init__(self, folder_dir, transform=None):
        # 1. Collect all images from both subfolders
        self.fpaths = glob(f"{folder_dir}/faulty/*.jpeg") + glob(f"{folder_dir}/ok/*.jpeg")

        # 2. Shuffle once with a fixed seed for reproducibility
        random.seed(42)
        random.shuffle(self.fpaths)

        self.transform = transform

    def __len__(self):
        return len(self.fpaths)

    def __getitem__(self, idx):
        fpath = self.fpaths[idx]
        img = read_image(fpath, mode=ImageReadMode.GRAY).float() / 255.0

        if self.transform:
            img = self.transform(img)

    # Use "in" to check the path string directly
        if "ok" in fpath.lower():
            label = 0.0
        elif "faulty" in fpath.lower(): # matches 'defect' or 'defective'
            label = 1.0
        else:
        # If this happens, your glob found a file that isn't in a labeled folder
            raise ValueError(f"Could not determine label for: {fpath}")

        return img, torch.tensor([label])


```

```python
import torchvision.transforms as T

IM_SIZE = 224

# Training: Focus on variety to prevent overfitting
```

```
train_tfms = T.Compose([
    T.Resize((IM_SIZE, IM_SIZE)),
    T.RandomHorizontalFlip(),
    T.RandomVerticalFlip(),
    T.RandomRotation(15),
    T.RandomResizedCrop(IM_SIZE, scale=(0.9, 1.0)),
    T.ColorJitter(brightness=0.1, contrast=0.1),
    T.Normalize(mean=[0.5], std=[0.5]),
])

# Testing: Focus on consistency for accurate evaluation
test_tfms = T.Compose([
    T.Resize((IM_SIZE, IM_SIZE)),
    T.CenterCrop(IM_SIZE),
    T.Normalize(mean=[0.5], std=[0.5]),
])
```

Model, Loss Function, Optimizer

```
import torch
import torch.nn as nn

class SimpleCNN(nn.Module):
    def __init__(self):
        super().__init__()

        # 1. Feature Extraction (Convolutional Layers)
        self.features = nn.Sequential(
            self._conv_block(1, 16),   # Output: 16 x 112 x 112
            self._conv_block(16, 32),  # Output: 32 x 56 x 56
            self._conv_block(32, 64),  # Output: 64 x 28 x 28
        )

        # 2. Classifier (Fully Connected Layers)
        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Linear(64 * 28 * 28, 128),
            nn.ReLU(inplace=True),
            nn.Dropout(0.5),
            nn.Linear(128, 1),
            nn.Sigmoid()
        )

        self.loss_fn = nn.BCELoss()

    def _conv_block(self, ni, no):
        """Helper to create a standard Conv-ReLU-BatchNorm-Pool block"""
        return nn.Sequential(
            nn.Conv2d(ni, no, kernel_size=3, padding=1),
```

```
            nn.ReLU(inplace=True),
            nn.BatchNorm2d(no),
            nn.MaxPool2d(2)
        )

    def forward(self, x):
        return self.classifier(self.features(x))

```

MLflow Setup

```
import mlflow
import mlflow.pytorch

mlflow.end_run()
!pkill -f mlflow
# 1. Kill any existing MLflow processes
!pkill -f mlflow

# 2. Restart MLflow with specific flags for tunnels
import subprocess
import time

# We add '*' to allowed-hosts to accept any tunnel URL
# We add '*' to cors-origins to allow the browser to render the UI
cmd = [
    'mlflow', 'ui',
    '--port', '5000',
    '--host', '0.0.0.0',
    '--allowed-hosts', '*',
    '--cors-allowed-origins', '*'
]

subprocess.Popen(cmd)

# Give the server 10 seconds to fully load its database
print("MLflow server is starting with security bypass...")
time.sleep(10)
```

```
MLflow server is starting with security bypass...
```

```
# Set the experiment name
mlflow.set_experiment("Casting_Defect_SimpleCNN")
```

```
<Experiment: artifact_location='/content/mlruns/1', creation_time=1771944911931, experiment_id='1', last_update_time=1771944911931, lifecycle_stage='active',
name='Casting_Defect_SimpleCNN', tags={}, workspace='default'>
```

```python
import torch
from torchsummary import summary

# Set Hardware Accelerator
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")

# Initialize and Move Model
model = SimpleCNN().to(device)

# Display Blueprint
# use (1, 224, 224) because the model is built for Grayscale (1 channel)
summary(model, input_size=(1, IM_SIZE, IM_SIZE))
```

```
Using device: cuda
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1         [-1, 16, 224, 224]             160
              ReLU-2         [-1, 16, 224, 224]               0
       BatchNorm2d-3         [-1, 16, 224, 224]              32
         MaxPool2d-4         [-1, 16, 112, 112]               0
            Conv2d-5         [-1, 32, 112, 112]           4,640
              ReLU-6         [-1, 32, 112, 112]               0
       BatchNorm2d-7         [-1, 32, 112, 112]              64
         MaxPool2d-8           [-1, 32, 56, 56]               0
            Conv2d-9           [-1, 64, 56, 56]          18,496
             ReLU-10           [-1, 64, 56, 56]               0
      BatchNorm2d-11           [-1, 64, 56, 56]             128
        MaxPool2d-12           [-1, 64, 28, 28]               0
          Flatten-13                [-1, 50176]               0
           Linear-14                  [-1, 128]       6,422,656
             ReLU-15                  [-1, 128]               0
          Dropout-16                  [-1, 128]               0
           Linear-17                    [-1, 1]             129
          Sigmoid-18                    [-1, 1]               0
================================================================
Total params: 6,446,305
Trainable params: 6,446,305
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.19
Forward/backward pass size (MB): 35.22
Params size (MB): 24.59
Estimated Total Size (MB): 60.00
----------------------------------------------------------------
```

```python
train_ds = CastingDataset('dataset/train', transform=train_tfms)
test_ds  = CastingDataset('dataset/test',  transform=test_tfms)

# 2. Initialize DataLoaders
# Dataset returns tensors directly
```

```
train_dl = DataLoader(train_ds, batch_size=32, shuffle=True)
test_dl  = DataLoader(test_ds,  batch_size=32, shuffle=False) # No need to shuffle test data

# 3. Inspect a single batch
images, labels = next(iter(train_dl))

print(f"Images batch shape: {images.shape}") # Expected: [32, 1, 224, 224]
print(f"Labels batch shape: {labels.shape}") # Expected: [32, 1]
```

```
Images batch shape: torch.Size([32, 1, 224, 224])
Labels batch shape: torch.Size([32, 1])
```

Validation step for Training/Testing batch

```
import torch.nn as nn
import pandas as pd
import torch
from torch.optim import Adam
from torch.utils.data import Dataset, DataLoader

# validation step for Training batch
def train_batch(images, labels, model, opt, loss_fn):
    model.train()
    preds = model(images)

    loss = loss_fn(preds, labels)
    acc = (preds > 0.5).float().eq(labels).float().mean()

    loss.backward()
    opt.step()
    opt.zero_grad()

    return loss.item(), acc.item()

# validation step for Testing batch
@torch.no_grad()
def validate_batch(images, labels, model, loss_fn):
    model.eval()
    preds = model(images)

    loss = loss_fn(preds, labels)
    acc = (preds > 0.5).float().eq(labels).float().mean()

    return loss.item(), acc.item()

```

Training Loop

```
# del log
# best_accuracy = 0.0
```

```
import os
import torch
from torch.optim import Adam
from torch_snippets import Report

# 1. Setup Models and directory
model = SimpleCNN().to(device)
optimizer = Adam(model.parameters(), lr=1e-3)
loss_fn = torch.nn.BCELoss()  # This replaces model.criterion
n_epochs = 20

model_dir = "/content/SimpleCNN/models"
os.makedirs(model_dir, exist_ok=True)

# 2. Training Initialization
log = Report(n_epochs)
best_accuracy = 0.0

# # Safety check: end any previous runs that might be hanging
# mlflow.end_run()

# # Now start your experiment
# mlflow.set_experiment("Casting_Defect_SimpleCNN")

# --- Start the MLflow Run ---
# New Hyper-parameter tuning
with mlflow.start_run(run_name="FineTuned_SimpleCNN_v2"):
    # Log the strategy
    mlflow.log_params({
        "lr": 1e-4,
        "pos_weight": 3.0,
        "dropout": 0.3,
        "epochs": n_epochs,
        "batch_size": 32,
        "optimizer": "Adam",
        "im_size": 128
    })

# # --- Start the MLflow Run ---
# # Existing Hyper-parameter
# with mlflow.start_run(run_name="SimpleCNN_Standard"):

#     # Log Hyperparameters
#     mlflow.log_params({
#         "lr": 1e-3,
#         "epochs": n_epochs,
```

```
#         "batch_size": 32,
#         "optimizer": "Adam",
#         "im_size": IM_SIZE
#     })

    for ex in range(n_epochs):
        # --- TRAINING PHASE ---
        N = len(train_dl)
        for bx, (imgs, lbls) in enumerate(train_dl):
            imgs, lbls = imgs.to(device), lbls.to(device)
            loss, acc = train_batch(imgs, lbls, model, optimizer, loss_fn)
            log.record(ex + (bx + 1)/N, trn_loss=loss, trn_acc=acc, end='\r')

        # --- VALIDATION PHASE ---
        N = len(test_dl)
        epoch_val_accs = []
        epoch_val_losses = []  # <--- FIX 1: Define the missing list

        for bx, (imgs, lbls) in enumerate(test_dl):
            imgs, lbls = imgs.to(device), lbls.to(device)
            val_loss, val_acc = validate_batch(imgs, lbls, model, loss_fn)

            log.record(ex + (bx + 1)/N, val_loss=val_loss, val_acc=val_acc, end='\r')
            epoch_val_accs.append(val_acc)
            epoch_val_losses.append(val_loss)


        # Calculate Averages
        current_val_acc = sum(epoch_val_accs) / len(epoch_val_accs)
        current_val_loss = sum(epoch_val_losses) / len(epoch_val_losses)

        # Log to MLflow
        mlflow.log_metric("trn_loss", loss, step=ex)
        mlflow.log_metric("trn_acc", acc, step=ex)
        mlflow.log_metric("val_acc", current_val_acc, step=ex)
        mlflow.log_metric("val_loss", current_val_loss, step=ex)

        # Report Averages and Save
        log.report_avgs(ex + 1)

        if current_val_acc > best_accuracy:
            best_accuracy = current_val_acc
            save_path = f"{model_dir}/best_casting_model.pth"
            torch.save(model.state_dict(), save_path)
            print(f"New Best Model Saved! Manual Avg Val Acc: {current_val_acc:.4f}")

        torch.save(model.state_dict(), f"{model_dir}/model_epoch_{ex+1}.pth")

```

```
EPOCH: 1.000  val_loss: 0.669  trn_loss: 5.344  trn_acc: 0.697  val_acc: 0.738  (6.19s - 117.66s remaining)
New Best Model Saved! Manual Avg Val Acc: 0.7229
EPOCH: 2.000  val_loss: 0.499  trn_loss: 0.647  trn_acc: 0.703  val_acc: 0.768  (12.07s - 108.59s remaining)
New Best Model Saved! Manual Avg Val Acc: 0.7903
EPOCH: 3.000  val_loss: 0.419  trn_loss: 0.516  trn_acc: 0.724  val_acc: 0.808  (17.92s - 101.55s remaining)
New Best Model Saved! Manual Avg Val Acc: 0.8076
EPOCH: 4.000  val_loss: 0.430  trn_loss: 0.472  trn_acc: 0.738  val_acc: 0.815  (23.91s - 95.63s remaining)
New Best Model Saved! Manual Avg Val Acc: 0.8368
EPOCH: 5.000  val_loss: 0.435  trn_loss: 0.479  trn_acc: 0.739  val_acc: 0.833  (29.82s - 89.46s remaining)
EPOCH: 6.000  val_loss: 0.370  trn_loss: 0.476  trn_acc: 0.775  val_acc: 0.844  (35.65s - 83.17s remaining)
New Best Model Saved! Manual Avg Val Acc: 0.8438
EPOCH: 7.000  val_loss: 0.367  trn_loss: 0.431  trn_acc: 0.766  val_acc: 0.826  (41.54s - 77.15s remaining)
EPOCH: 8.000  val_loss: 0.342  trn_loss: 0.451  trn_acc: 0.760  val_acc: 0.878  (47.42s - 71.14s remaining)
New Best Model Saved! Manual Avg Val Acc: 0.8785
EPOCH: 9.000  val_loss: 0.278  trn_loss: 0.409  trn_acc: 0.808  val_acc: 0.899  (53.38s - 65.24s remaining)
New Best Model Saved! Manual Avg Val Acc: 0.8993
EPOCH: 10.000  val_loss: 0.296  trn_loss: 0.402  trn_acc: 0.809  val_acc: 0.872  (59.30s - 59.30s remaining)
EPOCH: 11.000  val_loss: 0.316  trn_loss: 0.374  trn_acc: 0.821  val_acc: 0.899  (65.11s - 53.27s remaining)
EPOCH: 12.000  val_loss: 0.246  trn_loss: 0.386  trn_acc: 0.831  val_acc: 0.910  (70.91s - 47.28s remaining)
New Best Model Saved! Manual Avg Val Acc: 0.9097
EPOCH: 13.000  val_loss: 0.228  trn_loss: 0.370  trn_acc: 0.832  val_acc: 0.906  (76.81s - 41.36s remaining)
EPOCH: 14.000  val_loss: 0.175  trn_loss: 0.330  trn_acc: 0.853  val_acc: 0.913  (82.64s - 35.42s remaining)
New Best Model Saved! Manual Avg Val Acc: 0.9132
EPOCH: 15.000  val_loss: 0.251  trn_loss: 0.371  trn_acc: 0.844  val_acc: 0.910  (88.61s - 29.54s remaining)
EPOCH: 16.000  val_loss: 0.196  trn_loss: 0.331  trn_acc: 0.855  val_acc: 0.924  (94.41s - 23.60s remaining)
New Best Model Saved! Manual Avg Val Acc: 0.9236
EPOCH: 17.000  val_loss: 0.212  trn_loss: 0.339  trn_acc: 0.859  val_acc: 0.896  (100.36s - 17.71s remaining)
EPOCH: 18.000  val_loss: 0.185  trn_loss: 0.322  trn_acc: 0.850  val_acc: 0.924  (106.20s - 11.80s remaining)
EPOCH: 19.000  val_loss: 0.138  trn_loss: 0.269  trn_acc: 0.865  val_acc: 0.934  (112.07s - 5.90s remaining)
New Best Model Saved! Manual Avg Val Acc: 0.9340
EPOCH: 20.000  val_loss: 0.163  trn_loss: 0.284  trn_acc: 0.879  val_acc: 0.913  (118.02s - 0.00s remaining)
```

```
# # Check if you have both 0s and 1s in your batch
# imgs, lbls = next(iter(train_dl))
# print(f"Labels in this batch: {lbls.view(-1)}")
# # If you only see 0s or only 1s, your dataset loading logic is broken.
```

```
Labels in this batch: tensor[32] x∈[0., 1.000] μ=0.656 σ=0.483
```

Metrics & Confusion Matrix

```
log.plot_epochs(["val_loss", "trn_loss"])
log.plot_epochs(["val_acc", "trn_acc"])
```

```
100%|██████████| 121/121 [00:00<00:00, 6648.73it/s]
```

In Epoch 19:

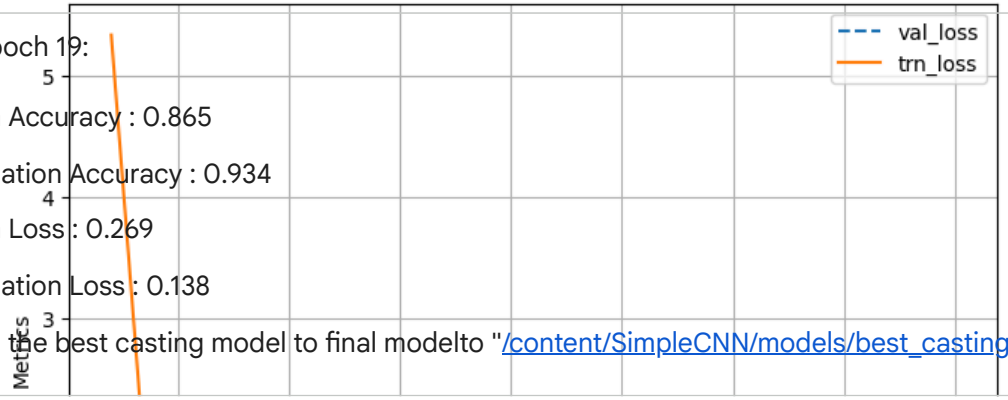
Train Accuracy : 0.865

Validation Accuracy : 0.934

Train Loss : 0.269

Validation Loss : 0.138

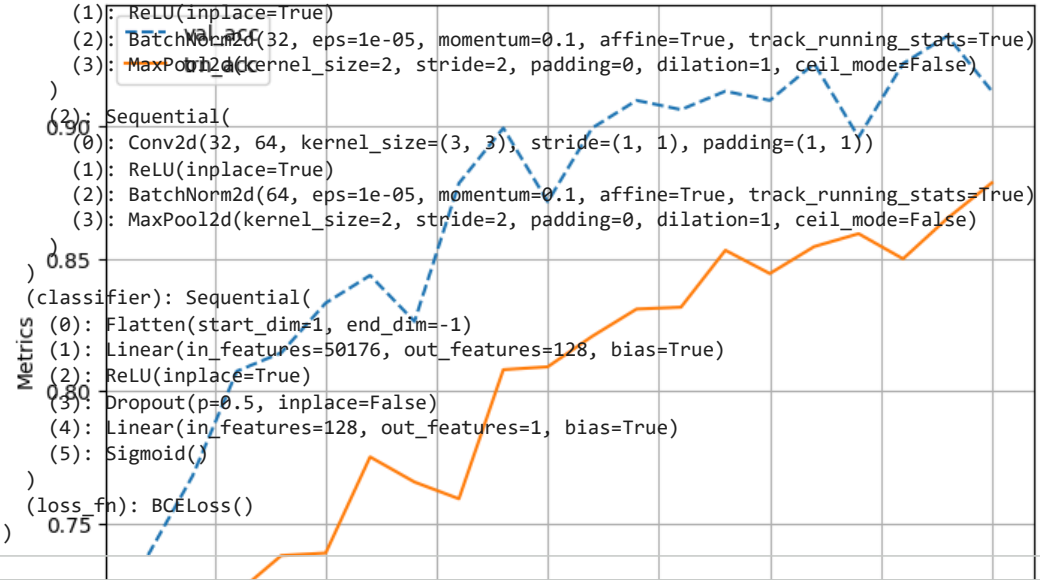
Save the best casting model to final modelto "/content/SimpleCNN/models/best_casting_model.pth"




```
final_model = SimpleCNN().to(device)
# Add 'models/' to the path
final_model.load_state_dict(torch.load("/content/SimpleCNN/models/best_casting_model.pth", weights_only=True))
final_model.eval() # Save the best model
```

```
SimpleCNN(
  (features): Sequential(
    (0): Sequential(
      (0): Conv2d(1, 16, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
      (1): ReLU(inplace=True)
      (2): BatchNorm2d(16, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (3): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
    )
    (1): Sequential(
      (0): Conv2d(16, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
      (1): ReLU(inplace=True)
      (2): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (3): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
    )
    (2): Sequential(
      (0): Conv2d(32, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
      (1): ReLU(inplace=True)
      (2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (3): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
    )
  )
  (classifier): Sequential(
    (0): Flatten(start_dim=1, end_dim=-1)
    (1): Linear(in_features=50176, out_features=128, bias=True)
    (2): ReLU(inplace=True)
    (3): Dropout(p=0.5, inplace=False)
    (4): Linear(in_features=128, out_features=1, bias=True)
    (5): Sigmoid()
  )
  (loss_fn): BCELoss()
)
```

```
100%|██████████| 121/121 [00:00<00:00, 6400.61it/s]
```



Predicting Batch

```python
@torch.no_grad()
def predict_batch(batch, model):
    model.eval()

    # 1. Unpack only two items (images and labels)
    images, labels = batch

    # 2. Move to device
    images = images.to(device)

    # 3. Get predictions from model
    preds = model(images)

    # 4. Convert probabilities to binary classes (0 or 1)
    # We use squeeze() to ensure [Batch, 1] becomes [Batch]
    preds_class = (preds.squeeze() > 0.5).long()

    # 5. Return everything on CPU for metrics/plotting
    # labels.squeeze() ensures the ground truth has the same shape as preds_class
    return preds_class.cpu(), labels.squeeze().cpu()
```

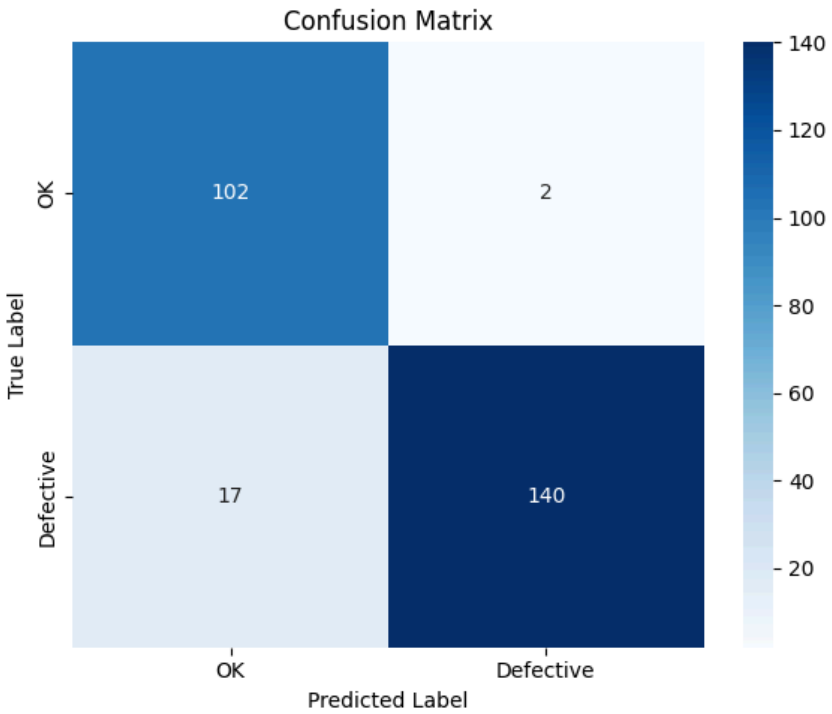
```python
all_preds = []
all_targets = []

for batch in test_dl:
    preds, targets = predict_batch(batch, final_model)
    all_preds.append(preds)
    all_targets.append(targets)

all_preds = torch.cat(all_preds)
all_targets = torch.cat(all_targets)

print("Predictions:", all_preds.shape)
print("Targets:", all_targets.shape)
```

```
Predictions: torch.Size([261])
Targets: torch.Size([261])
```

```python
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

cm = confusion_matrix(all_targets, all_preds)
print(cm)

plt.figure(figsize=(6, 5))
```

```
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["OK", "Defective"],
    yticklabels=["OK", "Defective"]
)

plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.title("Confusion Matrix")
plt.tight_layout()

# --- Save BEFORE Show ---
plt.savefig("cm_plot.png")
mlflow.log_artifact("cm_plot.png") # Uploads the image to the UI

plt.show()

# Extract counts from Confusion Matrix
tn, fp, fn, tp = cm.ravel()
mlflow.log_metric("false_negatives", fn)
mlflow.log_metric("false_positives", fp)

```

```
[[102   2]
 [ 17 140]]
```



f1, Precision and Recall Metrices

```
from sklearn.metrics import precision_score, recall_score, f1_score
import torch
import numpy as np

def get_metrics(model, dataloader, device):
    model.eval()
    all_preds = []
    all_labels = []

    with torch.no_grad():
        for x, y in dataloader: # Unpack 2 items only
            x, y = x.to(device), y.to(device)

            # 1. Get model output (already contains Sigmoid)
            outputs = model(x)

            # 2. Convert to binary classes
            # Squeeze ensures shape [32, 1] becomes [32]
```

```
            preds = (outputs.squeeze() > 0.5).int()

            all_preds.extend(preds.cpu().numpy())
            all_labels.extend(y.cpu().squeeze().numpy())

    # 3. Calculate metrics
    # We use 'binary' average by default
    f1 = f1_score(all_labels, all_preds)
    precision = precision_score(all_labels, all_preds)
    recall = recall_score(all_labels, all_preds)

    return f1, precision, recall

# Ensure to use if SimpleCNN instance
f1, precision, recall = get_metrics(model, test_dl, device)

print(f"Metrics on Test Set:")
print(f"F1-Score:  {f1:.4f}")
print(f"Precision: {precision:.4f}")
print(f"Recall:    {recall:.4f}")


# Log final calculation metrics
mlflow.log_metrics({
  "final_f1": f1,
  "final_precision": precision,
  "final_recall": recall
  })

# Save the final PyTorch model artifact
mlflow.pytorch.log_model(model, "simple_cnn_best_model")

# The 'with' block ends here
```

```
2026/02/24 18:05:56 WARNING mlflow.models.model: `artifact_path` is deprecated. Please use `name` instead.
2026/02/24 18:05:56 WARNING mlflow.pytorch: Saving pytorch model by Pickle or CloudPickle format requires exercising caution because these formats rely on Python's ob
2026/02/24 18:05:56 WARNING mlflow.utils.requirements_utils: Found torch version (2.9.0+cu126) contains a local version label (+cu126). MLflow logged a pip requiremen
Metrics on Test Set:
F1-Score:  0.9180
Precision: 0.9459
Recall:    0.8917
2026/02/24 18:06:01 WARNING mlflow.utils.requirements_utils: Found torch version (2.9.0+cu126) contains a local version label (+cu126). MLflow logged a pip requiremen
<mlflow.models.model.ModelInfo at 0x7e970ff2da60>
```

MLFlow tunnel activation

```
# 1. Kill any existing MLflow processes
!pkill -f mlflow
```

```python

# 2. Restart MLflow with specific flags for tunnels
import subprocess
import time

# add '*' to allowed-hosts to accept any tunnel URL
# add '*' to cors-origins to allow the browser to render the UI
cmd = [
    'mlflow', 'ui',
    '--port', '5000',
    '--host', '0.0.0.0',
    '--allowed-hosts', '*',
    '--cors-allowed-origins', '*'
]

subprocess.Popen(cmd)

# Give the server 10 seconds to fully load its database
print("MLflow server is starting with security bypass...")
time.sleep(10)
```

```
MLflow server is starting with security bypass...
```

```python
import getpass
import os

# Start the MLflow UI in the background on port 5000
get_ipython().system_raw("mlflow ui --port 5000 &")

# Get your public IP (this acts as the password for the localtunnel)
print("Your Tunnel Password (IP) is:")
!curl ipv4.icanhazip.com
# 1. Get your IP for the tunnel password
ip = !curl -s ipv4.icanhazip.com
print(f"Tunnel Password / Endpoint IP: {ip[0]}")

# 2. Start the tunnel
# We keep the local-host fix just in case
!npx localtunnel --port 5000 --local-host 127.0.0.1
```

```
Your Tunnel Password (IP) is:
35.247.179.234
Tunnel Password / Endpoint IP: 35.247.179.234
⠙⠹⠸⠼⠴⠦⠧⠇⠏⠋your url is: https://large-swans-poke.loca.lt
^C
```

2nd model - Pretrained Model + custom Classification Layer

Transfer Learning on VGG16

```python
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications.vgg16 import preprocess_input

# 1. Define Augmentation Settings
# Training gets augmentation; Test only gets preprocessing
train_datagen = ImageDataGenerator(
    preprocessing_function=preprocess_input,
    rotation_range=10,
    width_shift_range=0.1,
    height_shift_range=0.1,
    horizontal_flip=True
)

test_datagen = ImageDataGenerator(
    preprocessing_function=preprocess_input
)

# 2. Create Generators
# class_mode='binary' automatically maps folder names to 0 and 1 alphabetically
train_gen = train_datagen.flow_from_directory(
    "dataset/train",
    target_size=(224, 224),
    batch_size=32,
    class_mode="binary",
    classes=['ok', 'faulty'],
    color_mode="rgb"
)

test_gen = test_datagen.flow_from_directory(
    "dataset/test",
    target_size=(224, 224),
    batch_size=32,
    class_mode="binary",
    classes=['ok', 'faulty'],
    color_mode="rgb",
    shuffle=False
)

# Verify and Force Label Mapping
# Keras usually maps 'ok' to 0 and 'faulty' to 1 (alphabetical).
# We want to confirm or flip this so Defective = 1.
print(f"Current Mapping: {train_gen.class_indices}")

```

```
Found 1039 images belonging to 2 classes.
Found 261 images belonging to 2 classes.
Current Mapping: {'ok': 0, 'faulty': 1}
```

Load Pre-trained VGG16 Model

```
from tensorflow.keras.applications import VGG16

base_model = VGG16(
    weights="imagenet",
    include_top=False,
    input_shape=(224, 224, 3)
)
```

Freeze Pre-trained Layers

```
for layer in base_model.layers:
    layer.trainable = False
```

Add Custom Classification Head

```
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Flatten, Dense, Dropout

x = base_model.output
x = Flatten()(x)
x = Dense(256, activation="relu")(x)
x = Dropout(0.5)(x)
output = Dense(1, activation="sigmoid")(x)

modelvgg = Model(inputs=base_model.input, outputs=output)
```

Compile the Model

```
modelvgg.compile(
    optimizer="adam",
    loss="binary_crossentropy",
    metrics=["accuracy"]
)
```

Train the Model

```python
from tensorflow.keras.callbacks import ModelCheckpoint

# Define the "check point" for the model
checkpoint = ModelCheckpoint(
    'best_casting_model.h5',
    monitor='val_accuracy', # Watch validation accuracy
    verbose=1,
    save_best_only=True,    # Only save if val_accuracy improved
    mode='max'              # We want the maximum accuracy
)


# Pass the checkpoint into the fit function
history = modelvgg.fit(
    train_gen,
    epochs=20,
    validation_data=test_gen,
    callbacks=[checkpoint] # <--- It saves automatically now!
)

```

```
/usr/local/lib/python3.12/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121: UserWarning: Your `PyDataset` class should call `super().__init_
  self._warn_if_super_not_called()
Epoch 1/20
33/33 ━━━━━━━━━━━━━━━━━━━━ 0s 366ms/step - accuracy: 0.6506 - loss: 15.0961
Epoch 1: val_accuracy improved from -inf to 0.91571, saving model to best_casting_model.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend u
33/33 ━━━━━━━━━━━━━━━━━━━━ 18s 456ms/step - accuracy: 0.6527 - loss: 14.9368 - val_accuracy: 0.9157 - val_loss: 0.5508
Epoch 2/20
33/33 ━━━━━━━━━━━━━━━━━━━━ 0s 352ms/step - accuracy: 0.8818 - loss: 0.7780
Epoch 2: val_accuracy did not improve from 0.91571
33/33 ━━━━━━━━━━━━━━━━━━━━ 12s 374ms/step - accuracy: 0.8816 - loss: 0.7742 - val_accuracy: 0.8544 - val_loss: 0.3102
Epoch 3/20
33/33 ━━━━━━━━━━━━━━━━━━━━ 0s 363ms/step - accuracy: 0.8623 - loss: 0.3338
Epoch 3: val_accuracy improved from 0.91571 to 0.93103, saving model to best_casting_model.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend u
33/33 ━━━━━━━━━━━━━━━━━━━━ 13s 405ms/step - accuracy: 0.8626 - loss: 0.3326 - val_accuracy: 0.9310 - val_loss: 0.1626
Epoch 4/20
33/33 ━━━━━━━━━━━━━━━━━━━━ 0s 348ms/step - accuracy: 0.9026 - loss: 0.2407
Epoch 4: val_accuracy improved from 0.93103 to 0.93487, saving model to best_casting_model.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend u
33/33 ━━━━━━━━━━━━━━━━━━━━ 13s 390ms/step - accuracy: 0.9026 - loss: 0.2406 - val_accuracy: 0.9349 - val_loss: 0.1345
Epoch 5/20
33/33 ━━━━━━━━━━━━━━━━━━━━ 0s 350ms/step - accuracy: 0.9079 - loss: 0.2216
Epoch 5: val_accuracy improved from 0.93487 to 0.96169, saving model to best_casting_model.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend u
33/33 ━━━━━━━━━━━━━━━━━━━━ 13s 393ms/step - accuracy: 0.9086 - loss: 0.2201 - val_accuracy: 0.9617 - val_loss: 0.0921
Epoch 6/20
```

```
33/33 ━━━━━━━━━━━━━━━━━━━━ 0s 350ms/step - accuracy: 0.9042 - loss: 0.2155
Epoch 6: val_accuracy did not improve from 0.96169
33/33 ━━━━━━━━━━━━━━━━━━━━ 12s 372ms/step - accuracy: 0.9045 - loss: 0.2151 - val_accuracy: 0.9042 - val_loss: 0.2605
Epoch 7/20
33/33 ━━━━━━━━━━━━━━━━━━━━ 0s 349ms/step - accuracy: 0.9221 - loss: 0.2053
Epoch 7: val_accuracy improved from 0.96169 to 0.96552, saving model to best_casting_model.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend u
33/33 ━━━━━━━━━━━━━━━━━━━━ 13s 391ms/step - accuracy: 0.9221 - loss: 0.2049 - val_accuracy: 0.9655 - val_loss: 0.1003
Epoch 8/20
33/33 ━━━━━━━━━━━━━━━━━━━━ 0s 347ms/step - accuracy: 0.9472 - loss: 0.1277
Epoch 8: val_accuracy improved from 0.96552 to 0.98467, saving model to best_casting_model.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend u
33/33 ━━━━━━━━━━━━━━━━━━━━ 13s 390ms/step - accuracy: 0.9473 - loss: 0.1278 - val_accuracy: 0.9847 - val_loss: 0.0590
Epoch 9/20
33/33 ━━━━━━━━━━━━━━━━━━━━ 0s 354ms/step - accuracy: 0.9521 - loss: 0.1389
Epoch 9: val_accuracy did not improve from 0.98467
33/33 ━━━━━━━━━━━━━━━━━━━━ 12s 375ms/step - accuracy: 0.9516 - loss: 0.1398 - val_accuracy: 0.9540 - val_loss: 0.0962
Epoch 10/20
33/33 ━━━━━━━━━━━━━━━━━━━━ 0s 346ms/step - accuracy: 0.9489 - loss: 0.1328
Epoch 10: val_accuracy did not improve from 0.98467
33/33 ━━━━━━━━━━━━━━━━━━━━ 12s 368ms/step - accuracy: 0.9489 - loss: 0.1326 - val_accuracy: 0.9847 - val_loss: 0.0519
Epoch 11/20
33/33 ━━━━━━━━━━━━━━━━━━━━ 0s 351ms/step - accuracy: 0.9448 - loss: 0.1502
Epoch 11: val_accuracy improved from 0.98467 to 0.98851, saving model to best_casting_model.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend u
33/33 ━━━━━━━━━━━━━━━━━━━━ 13s 393ms/step - accuracy: 0.9448 - loss: 0.1501 - val_accuracy: 0.9885 - val_loss: 0.0413
Epoch 12/20
33/33 ━━━━━━━━━━━━━━━━━━━━ 0s 351ms/step - accuracy: 0.9688 - loss: 0.0973
Epoch 12: val_accuracy did not improve from 0.98851
33/33 ━━━━━━━━━━━━━━━━━━━━ 12s 373ms/step - accuracy: 0.9687 - loss: 0.0977 - val_accuracy: 0.9885 - val_loss: 0.0306
```

Model Evaluation

```python
import numpy as np

# Get ground truth labels
y_true = test_gen.classes

# Predict probabilities
y_prob = modelvgg.predict(test_gen)

# Convert probabilities to binary predictions
y_pred = (y_prob > 0.5).astype(int).ravel()
```

```
9/9 ━━━━━━━━━━━━━━━━━━━━ 2s 153ms/step
```

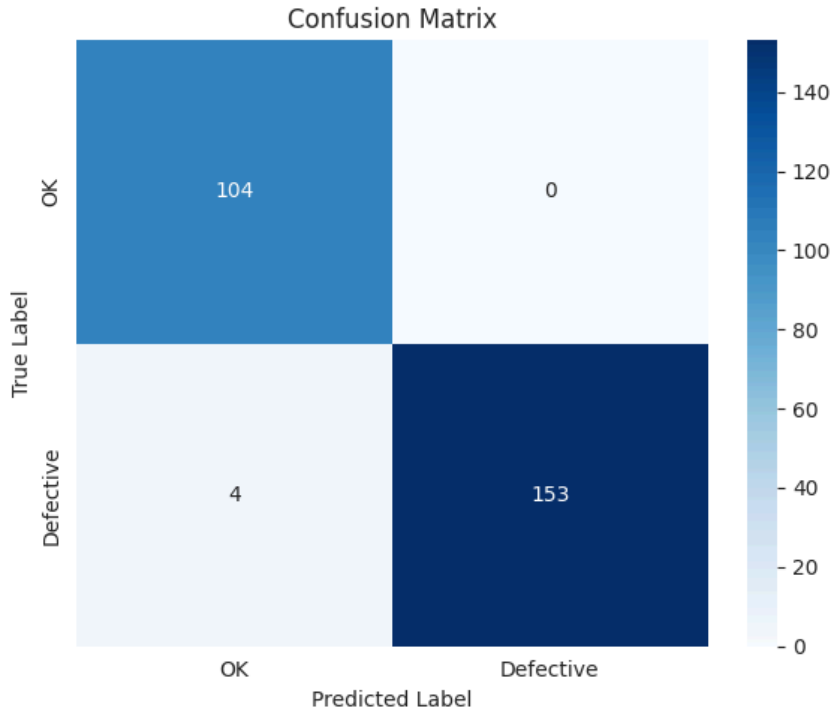
Precision, Recall, F1-score

```python
from sklearn.metrics import precision_score, recall_score, f1_score, classification_report

precision = precision_score(y_true, y_pred)
recall = recall_score(y_true, y_pred)
f1 = f1_score(y_true, y_pred)

print(f"Precision: {precision:.4f}")
print(f"Recall:    {recall:.4f}")
print(f"F1-score:  {f1:.4f}")
```

```
Precision: 1.0000
Recall:    0.9745
F1-score:  0.9871
```

Confusion Matrix

```python
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

cm = confusion_matrix(y_true, y_pred)

plt.figure(figsize=(6, 5))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["OK", "Defective"],
    yticklabels=["OK", "Defective"]
)

plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.title("Confusion Matrix")
plt.tight_layout()
plt.show()
```




```
import matplotlib.pyplot as plt

train_loss = history.history["loss"]
val_loss = history.history["val_loss"]
epochs = range(1, len(train_loss) + 1)

plt.figure(figsize=(8, 5))
plt.plot(epochs, train_loss, label="Training Loss")
plt.plot(epochs, val_loss, label="Validation Loss")

plt.title("Training vs Validation Loss (VGG16)")
plt.xlabel("Epochs")
plt.ylabel("Loss")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

train_acc = history.history["accuracy"]
val_acc = history.history["val_accuracy"]
epochs = range(1, len(train_acc) + 1)

```

```
plt.figure(figsize=(8, 5))
plt.plot(epochs, train_acc, label="Training Accuracy")
plt.plot(epochs, val_acc, label="Validation Accuracy")
```